

ANNU MATHUR

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EDUCATION

National Institute of Technology, Srinagar

B.Tech in Computer Science and Engineering | CGPA: 7.39

Aug 2023 – Aug 2027

Srinagar, India

WORK EXPERIENCE

Winter Research Intern, DIAT Pune (Govt. of India, MoD)

Dr. B.R. Ambedkar 3D AI Avatar (RAG System) | Guided by Prof. C.R.S Kumar

Dec 2025 – Jan 2026

[Live Demo](#) | [Certificate](#)

- Developed production-ready backend services for an AI-powered Retrieval-Augmented Generation application using FastAPI, Google Gemini, Python, and REST APIs.
- Implemented the complete RAG pipeline, including document ingestion, text chunking, embedding generation, semantic search, and vector database retrieval for context-aware AI responses.
- Designed API key failover, rate-limit handling, and error recovery mechanisms to improve reliability and ensure uninterrupted AI service availability during live demonstrations.
- Deployed the backend on Render and the frontend on Vercel, configured production environments, integrated MongoDB Atlas for query logging, and automated deployments using GitHub CI/CD.

PROJECTS

WanderNest – Vacation Rental Platform | Node.js, Express.js, MongoDB Atlas, Cloudinary, Passport.js

[GitHub](#)

- Developed a full-stack vacation rental platform with RESTful APIs supporting authentication, property listings, reviews, image uploads, and geolocation services.
- Implemented secure authentication and role-based authorization using Passport.js with MongoDB Atlas session management and encrypted user sessions.
- Designed scalable backend architecture using Express.js, integrated Cloudinary for image storage, and implemented robust server-side validation with Joi middleware.

Personal Portfolio | React.js, Node.js, Express.js, MongoDB, REST API

[GitHub](#)

- Built a MERN-stack portfolio with a REST API backend that serves project and skill data stored in MongoDB Atlas.
- Developed a responsive React frontend using reusable components and integrated it with REST APIs.

OPEN SOURCE CONTRIBUTION

Eventra – GirlScript Summer of Code 2026 | React.js, JavaScript, Git, GitHub

[Pull Request](#)

- Fixed responsive layout issues and refined form validation on the Feedback and Contact pages, removing duplicated UI components in the process.
- Resolved merge conflicts and addressed maintainer review feedback to get the change merged (PR #991).

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL

Backend: FastAPI, Node.js, Express.js, REST APIs

Frontend: React.js, HTML5, CSS3, Bootstrap

Databases: MongoDB Atlas, MySQL

Cloud & DevOps: Render, Vercel, GitHub Actions

Tools: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Cloudinary

Core Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

LEADERSHIP EXPERIENCE

Core Team Member — Range-Chinar Fest

Oct 2025 — [Certificate](#)

- Managed registration and payment verification for **1,000+ participants**, ensuring smooth check-in and accurate record management.
- Coordinated with volunteers and organizing teams to support seamless event execution and on-ground operations.

ACHIEVEMENTS & CERTIFICATIONS

- LeetCode:** 264+ problems solved, Contest Rating **1711** (Top 13.06%), Max Streak: 41
- DSA with Java** — Apna College
- Full-Stack Web Development** — Apna College
- Java (Basics)** — HackerRank

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